

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FIRST PARTIAL EXAMINATION - A.Y. 2023/2024

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Problem 2 - Eigenvalues Depending on a Free Parameter

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC]

Consider the LTI system

$$\begin{cases} \dot{x}(t) &= A x(t) + B u(t) \\ y(t) &= C x(t) + D u(t) \end{cases}$$

where the matrices are respectively:

$$A = \begin{bmatrix} 0 & -(80k + 15) \\ +1 & -2(8 + k) \end{bmatrix} \quad B = \begin{bmatrix} +80k \\ +2k \end{bmatrix} \quad C = [0 \quad +1] \quad D = [0]$$

and k is a strictly positive parameter: $k \in \mathbb{R}, k > 0$

Assignment:

- Determine the expression of the eigenvalues λ_1, λ_2 of the matrix A .
- Compute the eigenvalues when the gain k assumes the following values: $k = 0.1, 1, 100$
- Determine the characteristic polynomial $p_A(s, k)$ and of the matrix A .
- Compute the values of the parameter k that lead to coincident real eigenvalues. Once the values of k have been found, calculate the corresponding eigenvalues.
- Determine the value of the parameter k that leads to have the eigenvalues $\bar{\lambda}_{1,2} = -40 \pm j5 \sqrt{39}$.

Your code shall

- Define the **symbolic parameter** k and the **symbolic** complex **variable** s .
- Define the **symbolic matrix** A as in the above expression.
- Determine the expression of the eigenvalues of A . Sort the eigenvalues in ascending order and store those expressions in the **symbolic column vector** $\mathbf{eigA}$.
- Evaluate the expression of the eigenvalues of A when $k = 0.1, 1, 100$. Store the results in the **symbolic array** $\mathbf{eigSET}$. Each column of $\mathbf{eigSET}$ must correspond to a different value of the parameter k . Moreover, each column of $\mathbf{eigSET}$ has two elements.
- Determine the expression of the characteristic polynomial of the matrix A as a **symbolic function** of the complex variable s and the parameter k : $\mathbf{pA}(s, k)$.
- Compute the values of the parameter k that leads to real coincident eigenvalues for the matrix A .
- **Hint:** the characteristic polynomial $p_A(s, k)$ is a second-order polynomial, so to have two real and coincident roots, the **discriminant** of such a polynomial must be ...
- Sort the resulting values of k and store them in the symbolic column vector $\mathbf{k_sol}$.
- Once the values of $\mathbf{k_sol}$ have been found, calculate the corresponding eigenvalues, sort them and store them in the symbolic array $\mathbf{eig_sol}$. Each column of $\mathbf{eig_sol}$ corresponds to a different $\mathbf{k_sol}$ value.
- Determine the value of the parameter k that leads to the eigenvalues $\bar{\lambda}_{1,2} = -40 \pm j5 \sqrt{39}$.

- **Hint:** if $\bar{\lambda}_{1,2}$ are the actual eigenvalues, then a value $\hat{k}$ must exist, such that $p_A(\bar{\lambda}_1, \hat{k}) = 0$ and $p_A(\bar{\lambda}_2, \hat{k}) = 0$. Solving one of those equations allow us to find the desired value for k .
- Store the solution in the symbolic variable **k_d**.
- As validation of the finding, compute the eigenvalues of the matrix A when the parameter k assumes the value **k_d**, sort the eigenvalues and save them in the symbolic array **eigA_kd**.

Solution Proposal

```
syms s
syms k positive
syms A [2,2]
syms B [2, 1]
syms C [1, 2]
```

```
A = transpose([0, 1; -10*(1.5+8*k), -(16+2*k)])
```

```
A =
⎛ 0  -80 k - 15 ⎞
⎜ 1  -2 k - 16 ⎟
```

```
C = transpose(sym([0;1]))
```

```
c = (0 1)
```

```
B = transpose([80*k, 2*k])
```

```
B =
⎛ 80 k ⎞
⎜ 2 k ⎟
```

```
eigA = sort(eig(A), 'ascend')
```

```
eigA =
⎛ √k2 - 64 k + 49 - k - 8 ⎞
⎜ -k - √k2 - 64 k + 49 - 8 ⎟
```

```
kVALS = [0.1, 1, 100];
eigSET = sym(zeros(2, numel(kVALS)));

for n=1:numel(kVALS)
    dummyVAL = subs(eigA, k, kVALS(n));
    eigSET(:, n) = dummyVAL;
end
% eigSET = sort(eigSET, 'ascend');
pretty(eigSET)
```

```
/  sqrt(4261)  81
|  ----- - --,  - 9 + sqrt(14) 1i,  sqrt(3649) - 108
|      10      10
|
|  sqrt(4261)  81
|
|  \
```

$$\left| -\frac{\sqrt{10}}{10} - \frac{\sqrt{10}}{10}, -9 - \sqrt{14} \, i, -\sqrt{3649} - 108 \right|$$

```
pA(s, k) = charpoly(A,s)
```

$$pA(s, k) = s^2 + (2k + 16)s + 80k + 15$$

```
discriminant(k) = (2*k+16)^2-4*1*(80*k+15);
k_sol = sort(solve(discriminant==0), 'ascend')
```

$$k_sol = \begin{pmatrix} 32 - 5\sqrt{39} \\ 5\sqrt{39} + 32 \end{pmatrix}$$

```
k1_sol = k_sol(1); k2_sol = k_sol(2);
eigA1 = simplify(subs(eigA,k, k1_sol))
```

$$eigA1 = \begin{pmatrix} 5\sqrt{39} - 40 \\ 5\sqrt{39} - 40 \end{pmatrix}$$

```
eigA2 = simplify(subs(eigA, k, k2_sol))
```

$$eigA2 = \begin{pmatrix} -5\sqrt{39} - 40 \\ -5\sqrt{39} - 40 \end{pmatrix}$$

```
eig_sol = [eigA1, eigA2]
```

$$eig_sol = \begin{pmatrix} 5\sqrt{39} - 40 & -5\sqrt{39} - 40 \\ 5\sqrt{39} - 40 & -5\sqrt{39} - 40 \end{pmatrix}$$

```
eig_d = sym([-40-1i*5*sqrt(39); -40+1i*5*sqrt(39)]);
```

```
pA_kd(k) = collect(simplify(pA(eig_d(1),k)), k)
```

$$pA_kd(k) = (-10\sqrt{39} \, i)k + 320\sqrt{39} \, i$$

```
k_d = solve(pA_kd==0)
```

$$k_d = 32$$

```
eigA_kd = sort(simplify(subs(eigA,k, k_d)), 'ascend')
```

$$eigA_kd =$$

$$\begin{pmatrix} -40 - 5 \sqrt{39} \, i \\ -40 + \sqrt{975} \, i \end{pmatrix}$$